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(54) **SYSTEM AND METHOD FOR VERIFYING PRESENCE IN VIRTUAL ENVIRONMENTS**

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(52) **U.S. Cl.**

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See application file for complete search history.

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Primary Examiner — Kristie D Shingles

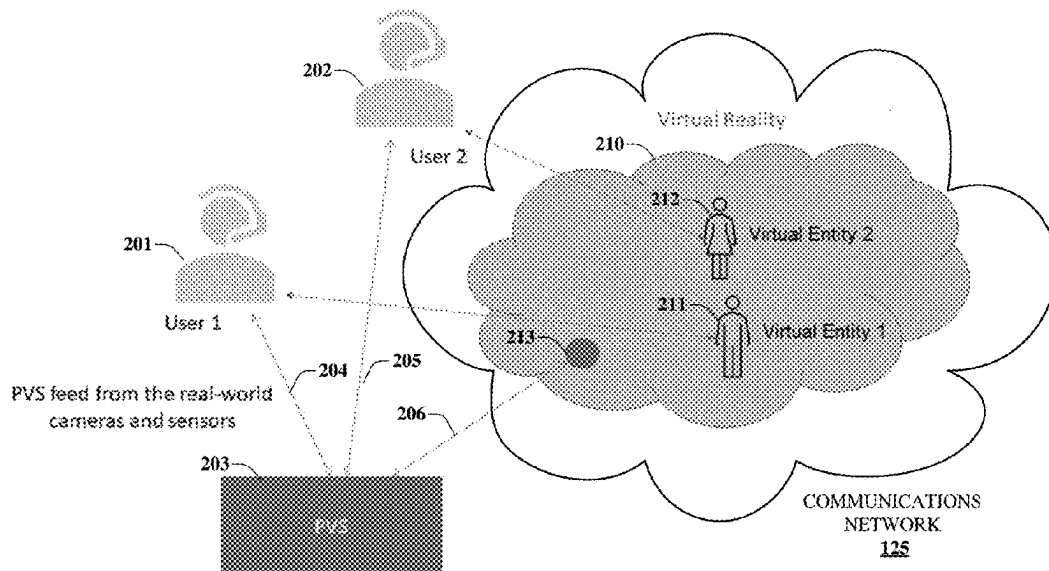
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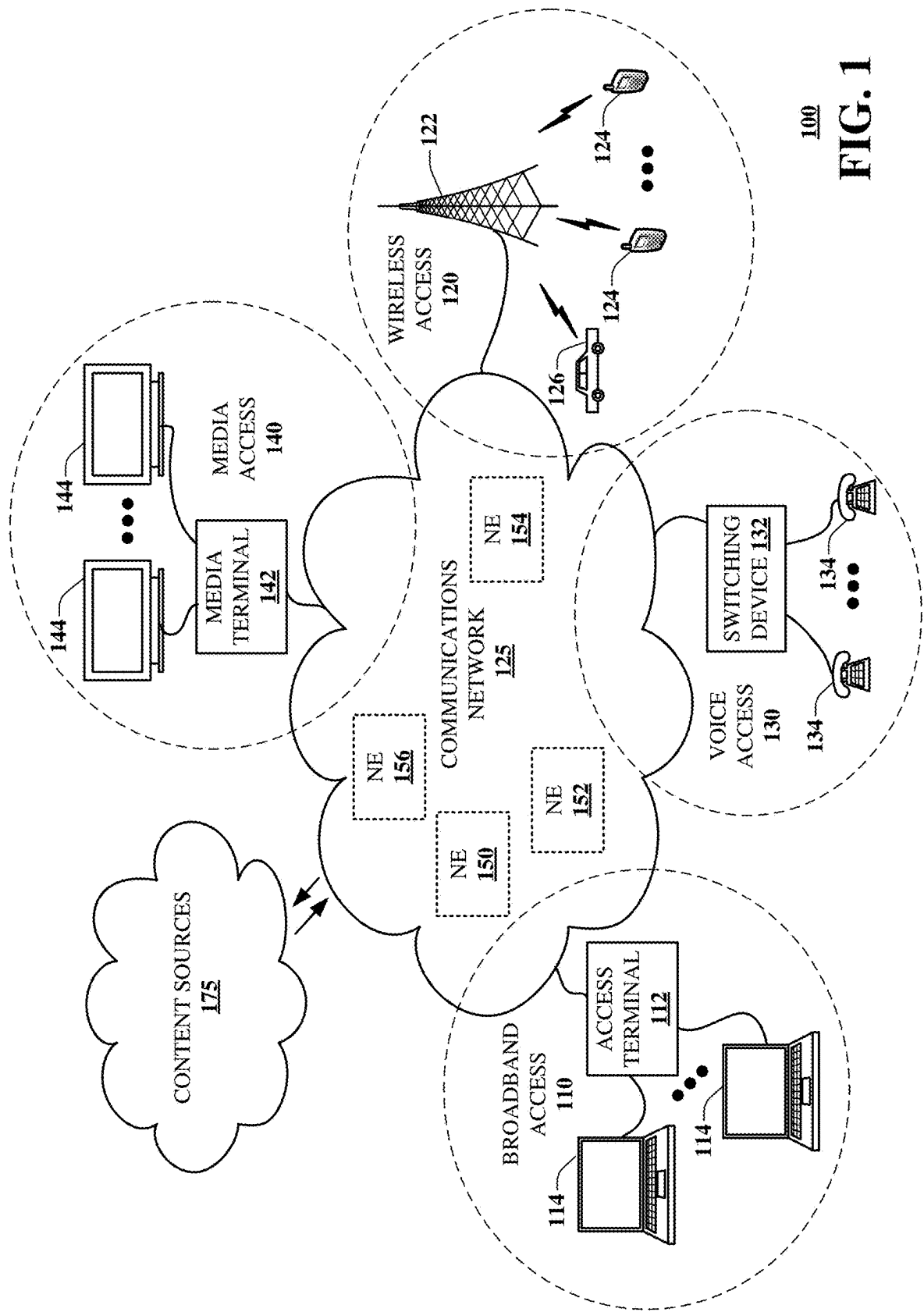
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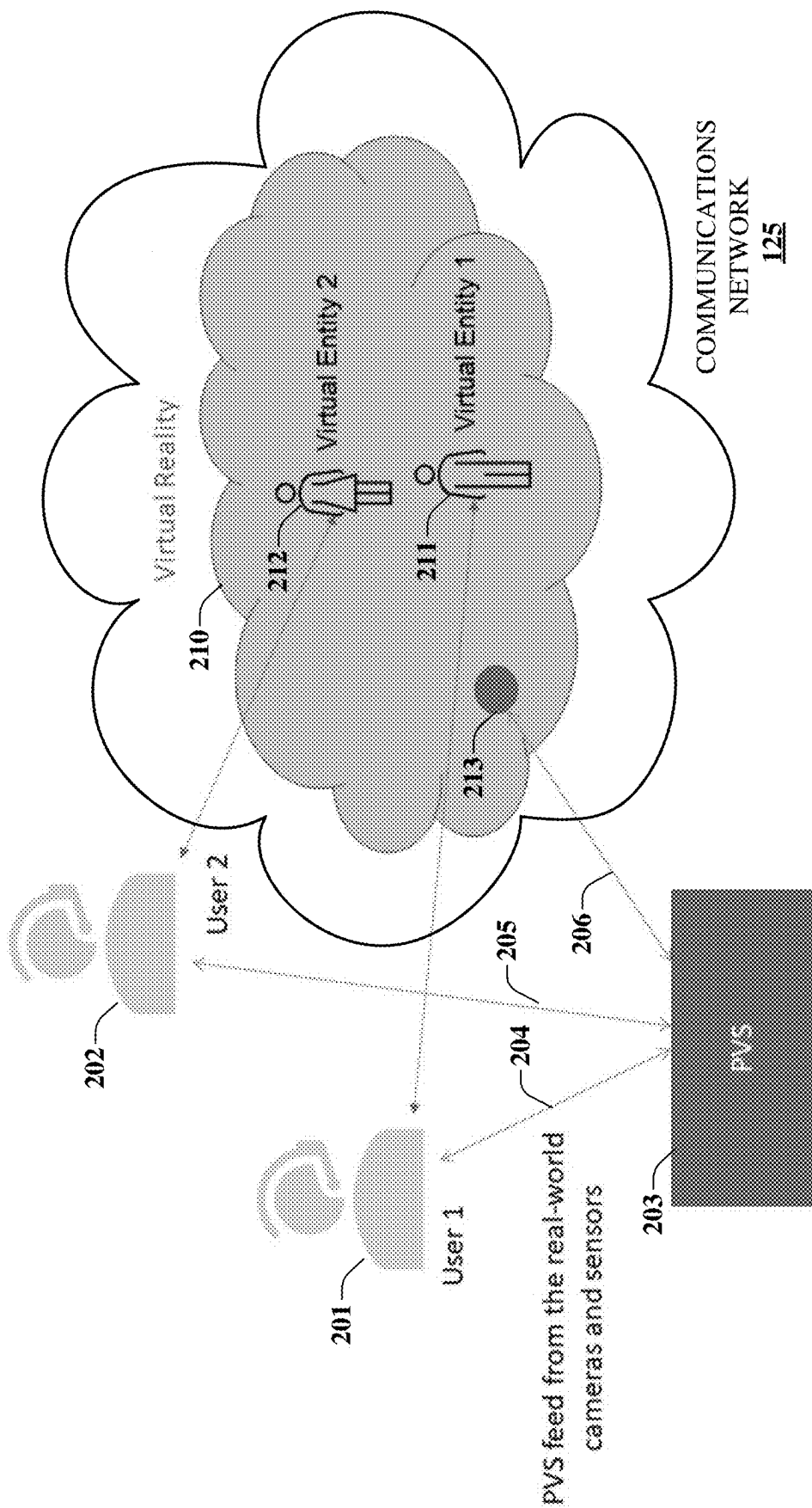
ABSTRACT

Aspects of the subject disclosure may include, for example, a device, including a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations of authenticating an identity of a user of user equipment that controls a virtual entity in a virtual environment; receiving a data feed providing images from the virtual environment; correlating signals from the user equipment with actions of the virtual entity observed in the images; and issuing a warning responsive to the user being unidentified or the signals being uncorrelated with the actions. Other embodiments are disclosed.

20 Claims, 7 Drawing Sheets

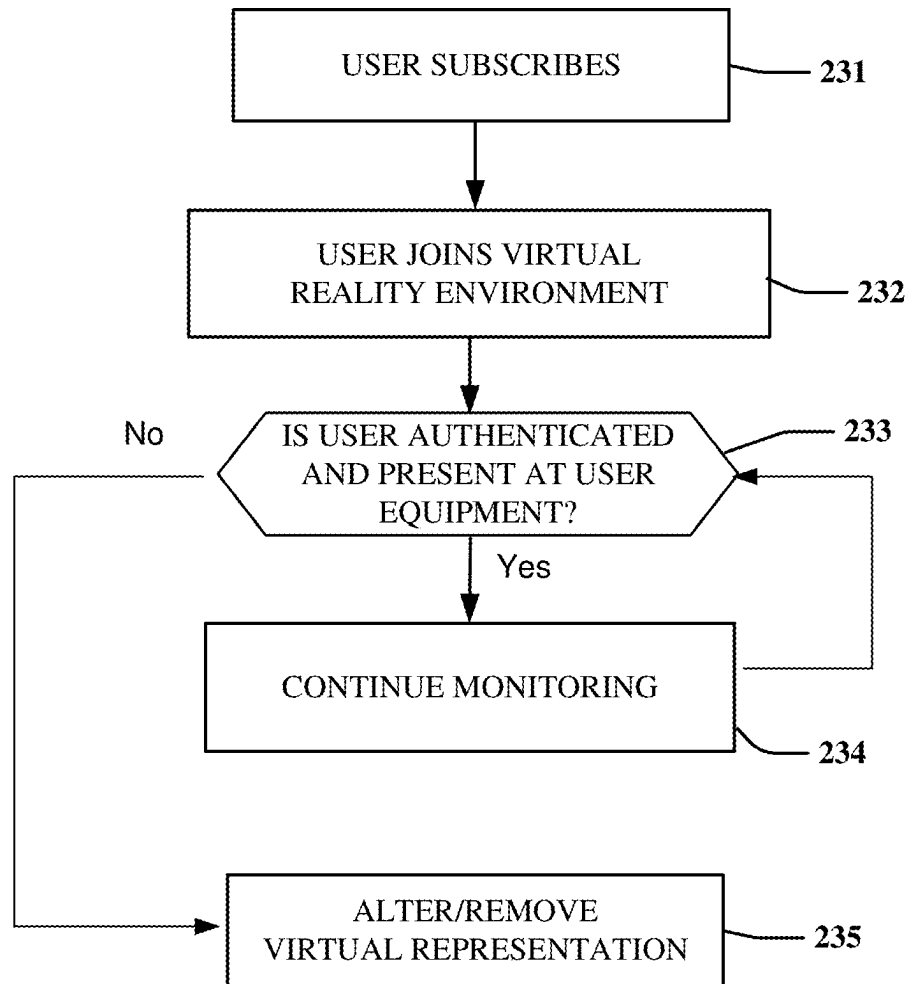


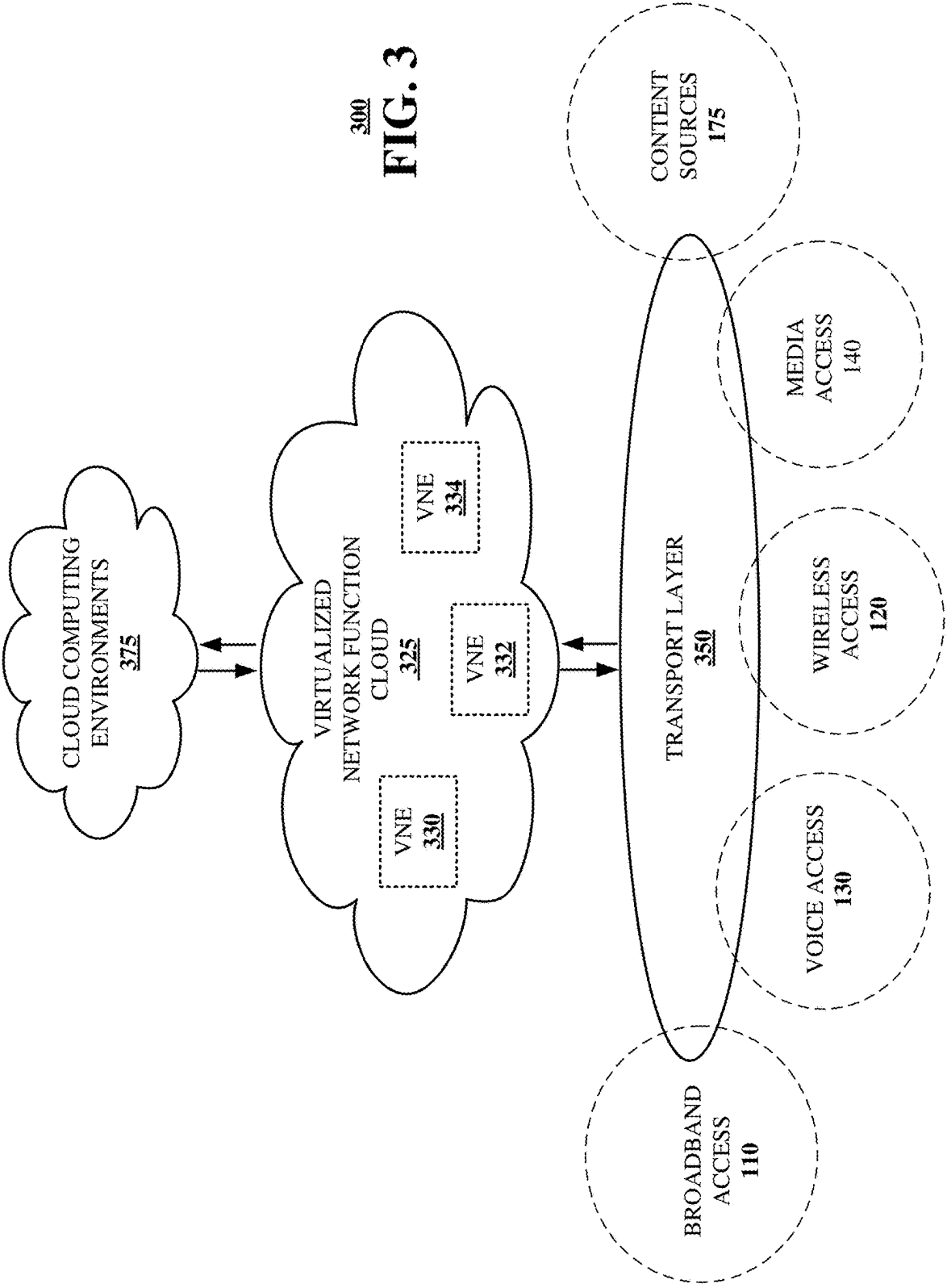




200

FIG. 2A

230**FIG. 2B**



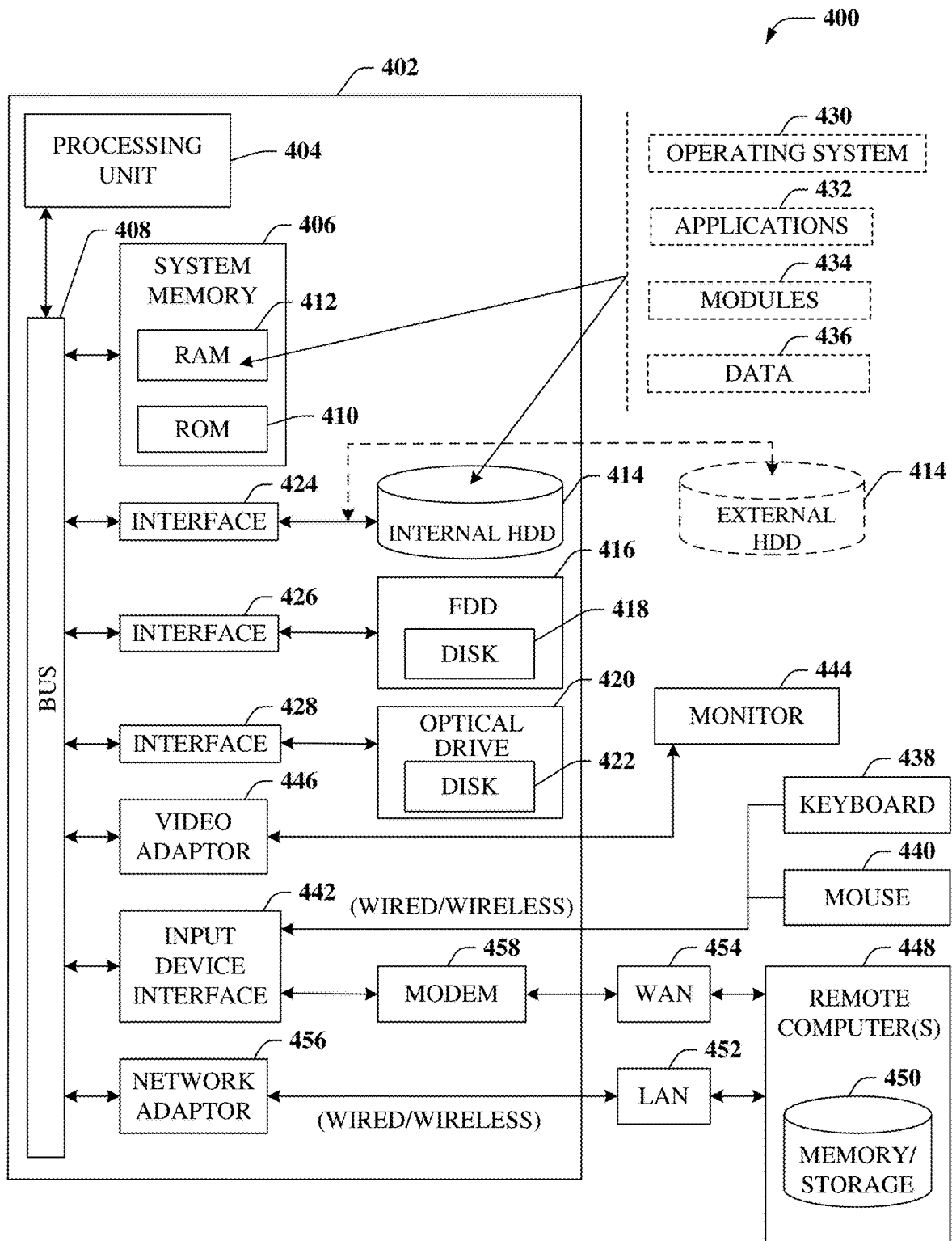


FIG. 4

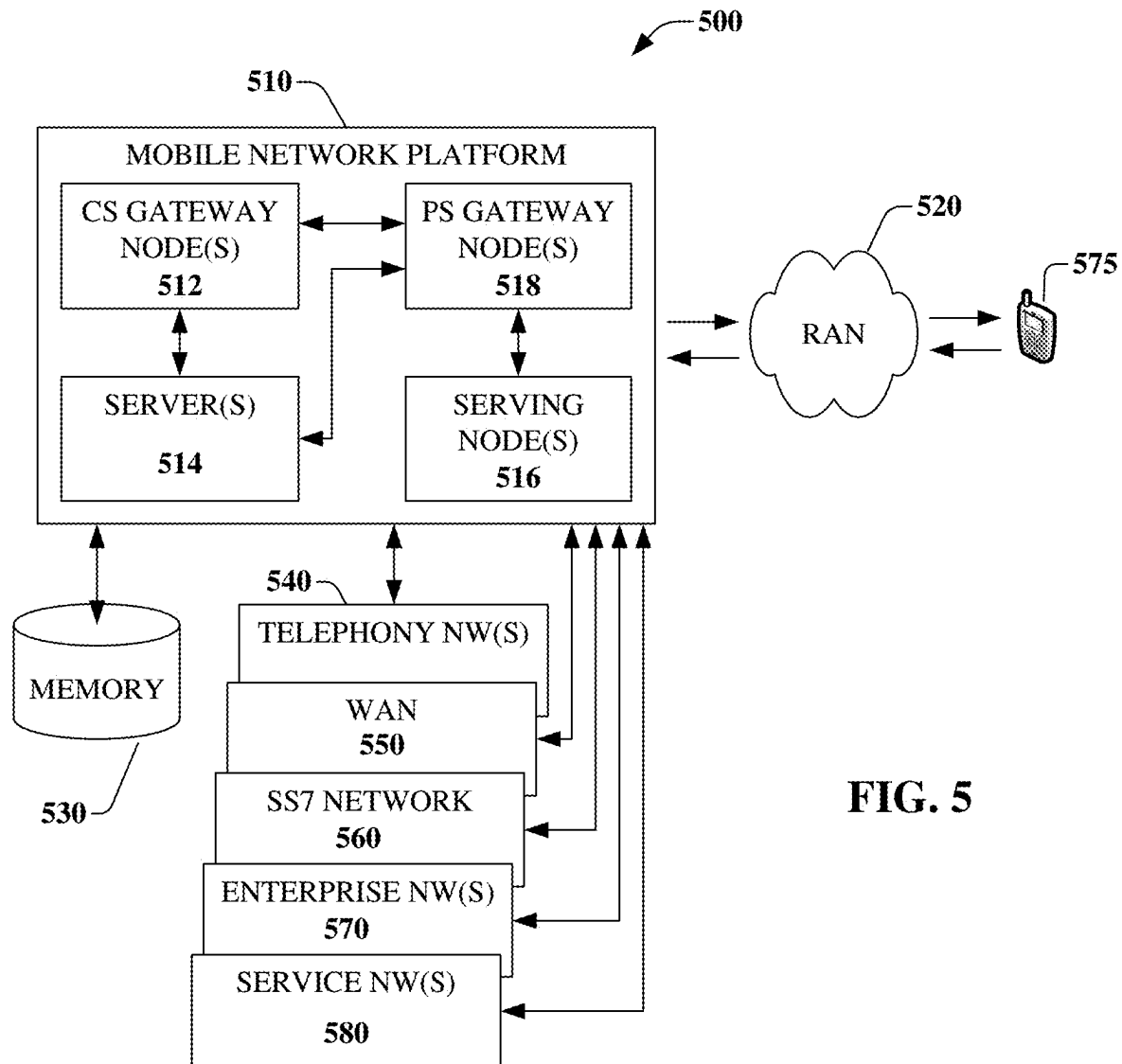
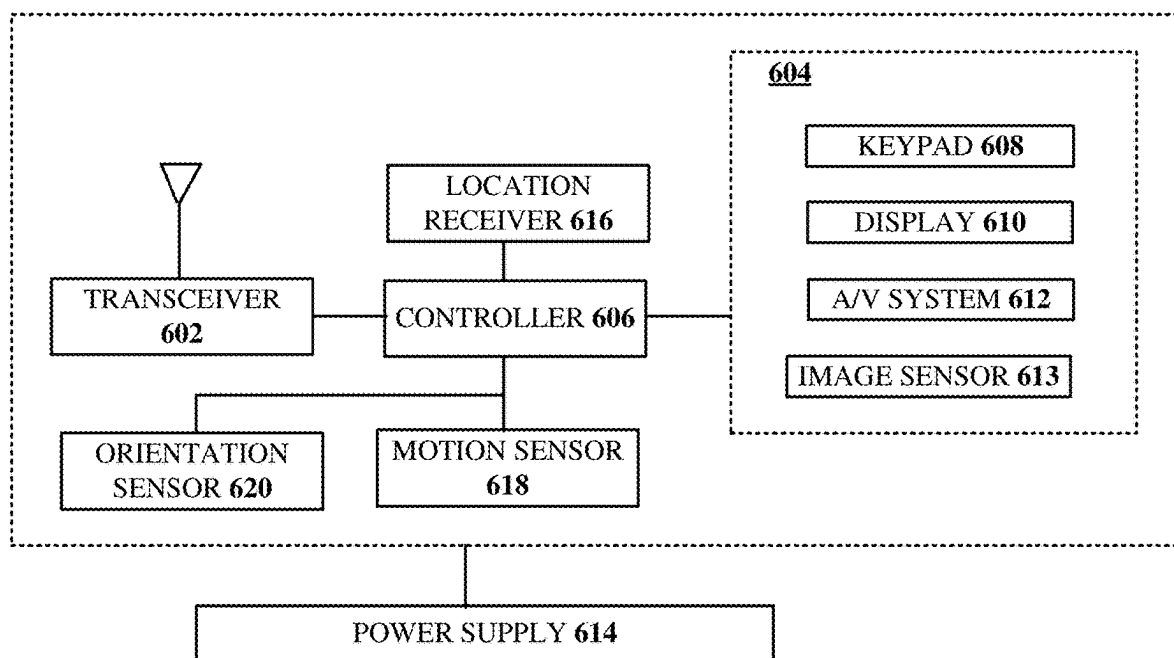


FIG. 5



600
FIG. 6

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SYSTEM AND METHOD FOR VERIFYING PRESENCE IN VIRTUAL ENVIRONMENTS

FIELD OF THE DISCLOSURE

The subject disclosure relates to a system and method for verifying presence in virtual environments.

BACKGROUND

Virtual reality environments are becoming mainstream tools for commerce. For example, the “Metaverse” is a fully immersive experience that can be an extension of the real world where users may strike deals and conduct important business.

BRIEF DESCRIPTION OF THE DRAWINGS

Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system for verifying presence in virtual environments, which functions within the communication network of FIG. 1 in accordance with various aspects described herein.

FIG. 2B depicts an illustrative embodiment of a method in accordance with various aspects described herein.

FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

The subject disclosure describes, among other things, illustrative embodiments for a system and method for verifying presence in virtual environments. Other embodiments are described in the subject disclosure.

One or more aspects of the subject disclosure include a device, including a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations of authenticating an identity of a user of user equipment that controls a virtual entity in a virtual environment; receiving a data feed providing images from the virtual environment; correlating signals from the user equipment with actions of the virtual entity observed in the images; and providing an indication responsive to the user being unidentified or the signals being uncorrelated with the actions.

One or more aspects of the subject disclosure include a non-transitory, machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations of authenticating an identity of a user of user equipment that controls a virtual entity in a virtual environ-

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ment; receiving a data feed providing images from the virtual environment; correlating signals from the user equipment with actions of the virtual entity observed in the images; continuously monitoring a presence of the user at the user equipment during a period of controlling the virtual entity; and providing an indication responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment.

One or more aspects of the subject disclosure include a method of receiving, by a processing system including a processor, a data feed providing audio and video from a virtual environment; correlating signals, by the processing system, from user equipment with actions of a virtual entity observed in the data feed; continuously monitoring, by the processing system, a presence of a user at the user equipment during a period of controlling the virtual entity; and issuing, by the processing system, a message responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment.

Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system 100 in accordance with various aspects described herein. For example, system 100 can facilitate in whole or in part providing a data feed of audio and video from a virtual environment; correlating signals from user equipment with actions of a virtual entity observed in the data feed; continuously monitoring a presence of a user at the user equipment during a period of controlling a virtual entity; and issuing a warning responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment. In particular, a communications network 125 is presented for providing broadband access 110 to a plurality of data terminals 114 via access terminal 112, wireless access 120 to a plurality of mobile devices 124 and vehicle 126 via base station or access point 122, voice access 130 to a plurality of telephony devices 134, via switching device 132 and/or media access 140 to a plurality of audio/video display devices 144 via media terminal 142. In addition, communication network 125 is coupled to one or more content sources 175 of audio, video, graphics, text and/or other media. While broadband access 110, wireless access 120, voice access 130 and media access 140 are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices 124 can receive media content via media terminal 142, data terminal 114 can be provided voice access via switching device 132, and so on).

The communications network 125 includes a plurality of network elements (NE) 150, 152, 154, 156, etc. for facilitating the broadband access 110, wireless access 120, voice access 130, media access 140 and/or the distribution of content from content sources 175. The communications network 125 can include a circuit switched or packet switched network, a voice over Internet protocol (VoIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

In various embodiments, the access terminal 112 can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals 114 can include personal computers, laptop

computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

In virtual reality environments, which includes augmented or mixed realities, the presence of an avatar or virtual entity does not necessarily indicate the identity of a real-world person represented by the virtual entity. In fact, the operation of some virtual entities may be purely autonomous, i.e., a mere manifestation of software running on a computer. The following disclosure illustrates a system and method to authenticate users as their virtual presence enters a virtual reality, also referred to as the “Metaverse,” but also authenticates their presence for the whole period that they are in the Metaverse. The system provides a “Proof of Presence” or continuous authentication that ensures only the authorized user is still in control of the virtual entity in the Metaverse, and therefore ensures that the virtual entity’s interactions commitments, speech, deeds, professional games, access private places, etc. are genuinely created by the real-world user.

The proposed system and method enable virtual reality users and people in the real-world as well to confidently rely on the virtual presence of a virtual entity as a representation of the actual, real-world person. For example, users may affirmatively realize that “Yes, John was in the virtual game/meeting/shopping/ . . . for the whole event yesterday

in the Metaverse.” The system provides authentication at the beginning of the interaction and continues the authentication during the interaction.

FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system for verifying presence in virtual environments, which functions within the communication network of FIG. 1 in accordance with various aspects described herein. As shown in FIG. 2A, system **200** comprises a presence verification system (PVS **203**), which may or may not be contained within the communications network **125**. In an embodiment, PVS **203** is implemented as a network function. Communications network **125** provides a virtual reality environment **210** to a first user equipment **201** and a second user equipment **202**. Virtual reality environment **210** manifests a first virtual entity **211** and a second virtual entity **212** correspondingly operated by first user equipment **201** and second user equipment **202** in the real-world, respectively. The first user equipment **201** and the second user equipment **202** comprises a head-mounted display, augmented reality glasses, a camera, a microphone, a joystick, a display screen, a keyboard, a mouse, a fingerprint reader, haptic sensors, or a combination thereof.

PVS **203** is a standalone system, separate from the virtual reality environment **210** that users may subscribe before participating in activities within the virtual reality environment **210**. PVS **203** comprises a data feed from the real-world environment of user **1** and user **2**. That is, PVS **203** includes a first data feed **204** from sensors at the first user equipment **201** and a second data feed **205** from sensors at the second user equipment **202**. Such sensors may include cameras, sensors, haptic devices, joysticks, or the like (not illustrated) that provide an indication of the presence and/or identity of user **1** or detect a lack of continuity of the operation of the respective user equipment.

PVS **203** further comprises a virtual camera **213** within virtual reality environment **210** that allows PVS **203** to observe first virtual entity **211** and second virtual entity **212**. This virtual data feed **206** from inside virtual reality environment **210** is like another invisible user placed inside the virtual space that provides audio and video from the virtual reality environment for monitoring the virtual entities inside that virtual space to PVS **203**. Virtual camera **213** monitors the behavior of first virtual entity **211** and second virtual entity **212** to ensure that real-world actions by the operators of first user equipment **201** and second user equipment **202** are correlated with the respective virtual entity’s actions. For example, PVS **203** observes the virtual entity’s behavior/movements/gestures/etc. and ensures that they match the inputs provided by first user equipment **201** and second user equipment **202**, respectively, so that continuity of operation is maintained and to prevent any man-in-the-middle spoofing of the operation of the virtual entities. In an embodiment, the virtual entity may be a holographic representation of the real-world user.

PVS **203** acts as a secondary security layer beyond any commercial authentication technique already provided for use of the user’s respective user equipment. Once a user is authenticated by PVS, if they wish to further verify their presence during their virtual interactions or transactions, then PVS **203** provides such verification of their presence. The cameras and sensors monitoring the physical space where the user equipment is located provide the following information to PVS **203**:

Face, facial expressions, etc.

body parts, body structure and expressions, body movements when controlling the user equipment (joystick, goggles, mouse, keyboard, etc.)

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The way the user holds the user equipment with their hand and fingers

Voice and vocal expressions

PVS 203 ensures the identity of the operator from these aspects. PVS 203 ensures that speech uttered by the operator in the physical world is the same speech heard in the virtual reality environment 210.

In an embodiment, if the identity of user 1 cannot be verified, for example by sensors at first user equipment 201, then PVS 203 will cause a noticeable change in the appearance of first virtual entity 211 within virtual reality environment 210. For example, second virtual entity 212 will see that first virtual entity 211 has a blurry image, for example, a blurred face. Other indications are contemplated within the scope of this disclosure, such as an adversarial image, like a red X over the user, or a warning sign indicating that this virtual entity is no longer authenticated. However, if PVS 203 is able to verify that user 1 is operating the first user equipment 201, then the image of first virtual entity 211 will appear clear and undistorted within virtual reality environment 210, as is normal.

In an embodiment, PVS 203 may develop a user profile with certain characteristics of the real-world users to further ensure that the identity of the real-world users is genuine. For example, biometric verification data of the user by sensors at the user equipment, such as facial recognition parameters, personal kinematics, biometric user interface analysis of input signal signatures, voice print analysis, two-factor authentication, keystroke analysis, mouse movements, or the like, may be stored in the profile for later comparison and authentication of the user.

In an exemplary scenario, two individual holograms meet in a virtual reality environment, with no physical contact. In this scenario, there are three spaces: two physical spaces for the operators and one virtual space where the holographic representations of the individual operators interact. PVS 203 has access to and monitors the three spaces to ensure that the actions of the operators correlate with the actions of the individual holograms.

In another exemplary scenario, two individuals meet in a hybrid space. A hybrid space is an augmented reality space where there are two physical spaces, one for each individual, which is augmented by a holographic representation of the individual that is not physically present in the space. Here, PVS 203 has access to both hybrid spaces and can ensure that the actions of each individual correlates with the actions of the individual holograms augmenting each physical space.

In an embodiment, entry to a virtual room will require subscribing to PVS 203. In another embodiment, a user may require another user to be subscribed to the PVS 203 before conducting certain virtual activities.

In an embodiment, virtual camera 213 is invisible to both virtual entities and merely observes their actions and interactions. In another embodiment, virtual camera 213 may be virtually represented as a virtual "eyewitness" that watches and records the actions of the parties in the virtual world, but otherwise does not interact with the virtual entities. In such case, the virtual presence of the virtual camera 213 may be virtually represented as an inanimate object, like a camera. After their interaction, PVS 203 can make the recording available to a subscriber of PVS 203.

In an embodiment, virtual camera 213 may have a virtual presence observable by the virtual entities and may step in to interrogate a virtual entity if there is a lack of correlation or authentication, to ensure that the genuine user is operating the virtual entity. The virtual presence may take on a

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humorous animated form, such as an owl, or may tell a joke, etc. and not interrupt the virtual entities, but wait for the opportune time to ask questions to confirm identity. Typical historical questions may be asked whose answers would only be known to the user, such as those used to verify identity by a credit reporting agency, for example. Upon successful interrogation, the virtual presence of virtual camera 213 may leave the virtual reality environment but will also remain invisibly present.

In yet another embodiment, virtual camera 213 may function as a virtual straw man that ensures performance of a virtual contract before delivering virtual objects from one virtual entity to another, essentially holding such objects in virtual escrow pending performance by one of the parties to the virtual contract.

FIG. 2B depicts an illustrative embodiment of a method in accordance with various aspects described herein. As shown in FIG. 2B, method 230 begins at step 231 where a user of the Virtual reality environment subscribes to the presence verification system. In step 232, the user joins the virtual reality environment. Then, in step 233, the system authenticates an identity of the user and checks that the user is present and operating the user equipment.

If the presence verification system can authenticate the identity of the user at the user equipment, then the process continues to step 234, where the presence verification system continuously monitors the real-world interactions with the user equipment and the resultant virtual reality actions by the corresponding virtual entity portrayed in the Virtual reality environment to ensure that there is a correlation between them. Periodically, the presence verification system may require authentication of the user at the user equipment as well.

If the presence verification system cannot authenticate the identity of the user at the user equipment, or the presence verification system detect a mismatch between real-world interactions with the user equipment and the resultant virtual reality actions by the corresponding virtual entity portrayed in the virtual reality environment, then in step 235 the presence verification system causes the virtual reality environment to provide a warning indication to other virtual entities in the virtual reality environment of an authentication/presence problem with the virtual entity. In an embodiment, the presence verification system may remove the virtual entity from the virtual reality environment or alter the appearance of the virtual entity to provide such indication.

While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2B, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

Referring now to FIG. 3, a block diagram 300 is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system 100, the subsystems and functions of system 200, and method 230 presented in FIGS. 1, 2A, 2B and 3. For example, virtualized communication network 300 can facilitate in whole or in part providing a data feed of audio and video from a virtual environment; correlating signals from user equipment with actions of a virtual entity observed in the data feed; continuously monitoring a presence of a user at

the user equipment during a period of controlling a virtual entity; and issuing a warning responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment.

In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general-purpose processors or general-purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it is elastic: so, the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized and might require special DSP code and analog front ends (AFE) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud opera-

tions, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements do not typically need to forward substantial amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and which creates an elastic function with higher availability overall than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud or might simply orchestrate workloads supported entirely in NFV infrastructure from these third-party locations.

Turning now to FIG. 4, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. 4 and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part providing a data feed of audio and video from a virtual environment; correlating signals from user equipment with actions of a virtual entity observed in the data feed; continuously monitoring a presence of a user at the user equipment during a period of controlling a virtual entity; and issuing a warning responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment.

Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer

electronics, and the like, each of which can be operatively coupled to one or more associated devices.

As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

With reference again to FIG. 4, the example environment can comprise a computer 402, the computer 402 comprising a processing unit 404, a system memory 406 and a system

bus 408. The system bus 408 couples system components including, but not limited to, the system memory 406 to the processing unit 404. The processing unit 404 can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit 404.

The system bus 408 can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures.

The system memory 406 comprises ROM 410 and RAM 412. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer 402, such as during startup. The RAM 412 can also comprise a high-speed RAM such as static RAM for caching data.

The computer 402 further comprises an internal hard disk drive (HDD) 414 (e.g., EIDE, SATA), which internal HDD 414 can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) 416, (e.g., to read from or write to a removable diskette 418) and an optical disk drive 420, (e.g., reading a CD-ROM disk 422 or, to read from or write to other high-capacity optical media such as the DVD). The HDD 414, magnetic FDD 416 and optical disk drive 420 can be connected to the system bus 408 by a hard disk drive interface 424, a magnetic disk drive interface 426 and an optical drive interface 428, respectively. The hard disk drive interface 424 for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer 402, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

A number of program modules can be stored in the drives and RAM 412, comprising an operating system 430, one or more application programs 432, other program modules 434 and program data 436. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM 412. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

A user can enter commands and information into the computer 402 through one or more wired/wireless input devices, e.g., a keyboard 438 and a pointing device, such as a mouse 440. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing

unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work,

without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part providing a data feed of audio and video from a virtual environment; correlating signals from user equipment with actions of a virtual entity observed in the data feed; continuously monitoring a presence of a user at the user equipment during a period of controlling a virtual entity; and issuing a warning responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment. In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, which facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN)), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology (ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be

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interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. 1(s) that enhance wireless service coverage by providing more network coverage.

It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

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In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part providing a data feed of audio and video from a virtual environment; correlating signals from user equipment with actions of a virtual entity observed in the data feed; continuously monitoring a presence of a user at the user equipment during a period of controlling a virtual entity; and issuing a warning responsive to the user being unidentified, the signals being uncorrelated with the actions or the user not being present at the user equipment.

The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, Wi-Fi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1x, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of

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the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, Wi-Fi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of

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flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

Other components not shown in FIG. 6 can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

The terms "first," "second," "third," and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, "a first determination," "a second determination," and "a third determination," does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

In the subject specification, terms such as "store," "storage," "data store," data storage," "database," and substantially any other information storage component relevant to operation and functionality of a component, refer to "memory components," or entities embodied in a "memory" or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, non-volatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-DRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

In one or more embodiments, information regarding use of services can be generated including services being

accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for conducting various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $x = (x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x) = \text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the

server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms represent-

ing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

As used herein, terms such as “data storage,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue”

indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

As may also be used herein, the term(s) “operably coupled to,” “coupled to,” and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

What is claimed is:

1. A device, comprising:

- a processing system including a processor; and
- a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:
 - authenticating an identity of a user of user equipment that controls a virtual entity in a virtual environment, wherein the virtual entity is a representation or avatar for the user in the virtual environment;
 - receiving a first data feed including signals from the user equipment, wherein the signals correspond to inputs to the user equipment that are associated with first actions in the virtual environment;
 - receiving a second data feed from a virtual camera providing images including the virtual entity performing second actions in the virtual environment;
 - responsive to the receiving of the first data feed and the second data feed, performing verification of the user by comparing the signals received from the user equip-

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ment with the second actions of the virtual entity presented in the images; and
based on determining the signals are uncorrelated with the second actions, causing the virtual environment to provide an indication that the signals are uncorrelated with the second actions.

2. The device of claim 1, wherein the operations further comprise continuously monitoring a presence of the user at the user equipment during a period of controlling the virtual entity.

3. The device of claim 2, wherein the identity of the user is authenticated by the user equipment.

4. The device of claim 3, wherein the data second feed further provides audio from the virtual environment and the authenticating comprises a comparison of biometric verification data with facial recognition, voice recognition, two-factor authentication, biometric user interface analysis, personal kinematics, or a combination thereof.

5. The device of claim 4, wherein the operations further comprise storing the biometric verification data in a user profile.

6. The device of claim 5, wherein the user equipment comprises a head-mounted display, augmented reality glasses, a camera, a microphone, a joystick, a display screen, a keyboard, a mouse, a fingerprint reader, haptic sensors, or a combination thereof.

7. The device of claim 6, wherein the indication comprises blurring the virtual entity or removing the virtual entity from the virtual environment.

8. The device of claim 7, wherein the operations are performed responsive to the virtual entity entering a virtual room.

9. The device of claim 8, wherein the operations further comprise:

representing the second data feed as an inanimate object in the virtual environment;
recording the images from the virtual environment; and
providing the recording to a subscriber.

10. The device of claim 8, wherein the operations further comprise: representing the second data feed as a second virtual entity in the virtual environment, and wherein the authenticating comprises controlling the second virtual entity to interrogate the virtual entity.

11. The device of claim 10, wherein the processing system comprises a plurality of processors operating in a distributed computing environment.

12. A non-transitory, machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising:

authenticating an identity of a user of user equipment that controls a virtual entity in a virtual environment, wherein the virtual entity is a representation or avatar for the user in the virtual environment;

receiving a first data feed including signals from the user equipment, wherein the signals correspond to inputs to the user equipment that are associated with first actions in the virtual environment;

receiving a second data feed from a virtual camera providing images including the virtual entity performing second actions in the virtual environment;

responsive to the receiving of the first data feed and the second data feed, performing verification of the user by comparing signals received from the user equipment with the second actions of the virtual entity observed in the images;

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continuously monitoring a presence of the user at the user equipment during a period of controlling the virtual entity; and

based on determining the signals are uncorrelated with the second actions, causing the virtual environment to provide an indication that the signals are uncorrelated with the second actions.

13. The non-transitory, machine-readable medium of claim 12, wherein the identity of the user is authenticated by the user equipment.

14. The non-transitory, machine-readable medium of claim 12, wherein the indication comprises blurring the virtual entity or removing the virtual entity from the virtual environment.

15. The non-transitory, machine-readable medium of claim 12, wherein the second data feed further provides audio from the virtual environment and the authenticating comprises a comparison of biometric verification data with facial recognition, voice recognition, two-factor authentication, biometric user interface analysis, personal kinematics or a combination thereof.

16. The non-transitory, machine-readable medium of claim 15, wherein the operations further comprise storing the biometric verification data in a user profile.

17. The non-transitory, machine-readable medium of claim 16, wherein the operations further comprise: representing the second data feed as a second virtual entity in the virtual environment, and wherein the authenticating comprises controlling the second virtual entity to interrogate the virtual entity.

18. The non-transitory, machine-readable medium of claim 12, wherein the processing system comprises a plurality of processors operating in a distributed computing environment.

19. A method, comprising:

receiving, by a processing system including a processor, a first data feed including signals from a user equipment, wherein the signals correspond to inputs to the user equipment that are associated with first actions in a virtual environment;

receiving, by the processing system, a second data feed from a virtual camera providing audio and video from a virtual environment;

responsive to the receiving of the first data feed and the second data feed, performing, by the processing system, verification of a user of the user equipment by comparing the signals received from the user equipment with second actions of a virtual entity presented in the video, wherein the virtual entity is a representation or avatar for the user in the virtual environment; continuously monitoring, by the processing system, a presence of a user at the user equipment during a period of controlling the virtual entity; and

based on determining the signals are uncorrelated with the second actions, issuing, by the processing system, a message responsive to the signals being uncorrelated with the second actions.

20. The method of claim 19, comprising:

representing, by the processing system, the second data feed as a second virtual entity in the virtual environment;

authenticating, by the processing system, an identity of the user by controlling the second virtual entity to interrogate the virtual entity; and

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issuing, by the processing system, the message responsive
to the user being unidentified by the authenticating.

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